

MA 102 (Mathematics II - Ordinary Differential Equations)

Department of Mathematics, IIT Guwahati

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Tutorial Sheet No. 1

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Lectures 1-4: Basics of ODEs, Formation of ODE, Separable ODE, Exact ODE, Integrating Factors, Special Nonlinear ODEs.

1. Determine the *order* and *degree* of the following ordinary differential equations. Also, identify these ODEs as *linear* or *nonlinear*.

(i) $\frac{d^2y}{dx^2} + x \sin y = 0$

(ii) $\frac{d^2y}{dx^2} + y \sin x = 0$

(iii) $\left(1 + \frac{dy}{dx}\right)^{\frac{1}{2}} = x \frac{d^2y}{dx^2}$

(iv) $\frac{d^6y}{dx^6} + \left(\frac{d^4y}{dx^4}\right) \left(\frac{d^3y}{dx^3}\right) + y = x$

(v) $\sin\left(\frac{d^3y}{dx^3}\right) = x + y.$

(vi) $\frac{d^3y}{dx^3} = \sin(x + y).$

(vii) $\frac{d^3y}{dx^3} + \sin\left(\frac{d^2y}{dx^2}\right) = x + y$

(viii) $\left(\frac{d^2y}{dx^2}\right)^2 + 4\frac{\left(\frac{d^2y}{dx^2}\right)^3}{\left(\frac{d^4y}{dx^4}\right)^2} + \frac{d^4y}{dx^4} = 0.$

2. Eliminating the arbitrary constants c_1 and c_2 , obtain the differential equation satisfied by the following functions.

(a) $y = c_1e^{-x} + c_2e^{2x}$

(b) $x^2 + c_1y^2 = 1$

(c) $y = c_1x - c_1^3$

3. Are the following differential equations exact? If so, obtain their general solutions.

(i) $(2xy - \sec^2 x)dx + (x^2 + 2y)dy = 0$

(ii) $(x - 2xy + e^y)dx + (y - x^2 + xe^y)dy = 0$

4. Solve the following ODEs:

- (i) $2x \, dx + 3(1 - y^3) \, dy = 0$
- (ii) $2xy \, dx + (x^2 + y^2) \, dy = 0$
- (iii) $x \sin y \, dx + (x + 1) \cos y \, dy = 0$
- (iv) $(y^2 + xy + x^2) \, dx - x^2 \, dy = 0$
- (v) $\frac{dy}{dx} - 2(\cos x)y = \cos x, y(0) = \frac{1}{2}$
- (vi) $\frac{dy}{dx} + 2y = e^x$

5. Find the values/conditions of the arbitrary constants/functions such that the following differential equations become exact, and hence solve them.

- (i) $(ax + by)dx + (cx + dy)dy = 0$
- (ii) $(x^2 + y^2)dx + (axy + y^4)dy = 0$
- (iii) $(2x + f(y))dx + 2xy \, dy = 0$

6. Consider the following differential equation.

$$\frac{dy}{dx} = f(x)g(y),$$

where f is a continuous function with $f(x) \neq 0$ and g is a continuously differentiable function. Then, prove that $y(x) = c$ is a constant solution of the given ODE if and only if $g(c) = 0$.

7. Show that the solution of $y' = k^2y$, $y(x_0) = y_0$ is increasing if $y_0 > 0$ and decreasing if $y_0 < 0$.

8. Consider the ODE $y' + ky = 1$, where k is a constant.

- (a) For what value of k will all solutions tend to 2 as $x \rightarrow +\infty$?
- (b) Is there any value of k for which there exists a non-constant solution $y(x)$ such that $y(x) \rightarrow -3$ as $x \rightarrow \infty$? Explain.

9. Solve the following Bernoulli's differential equations:

- (i) $x \frac{dy}{dx} - y = x^k y^n, \quad n \neq 1, k + n \neq 1$
- (ii) $2 \frac{dy}{dx} + y = (x - 1)y^3$
- (iii) $x^3 \frac{dy}{dx} - x^2 y + y^4 \cos x = 0$

10. Write the following differential equations in Clairaut's form:

- (i) $\frac{dy}{dx} = \ln \left(x \frac{dy}{dx} - y \right)$
- (ii) $\frac{dy}{dx} = \cos^{-1} \left(y - x \frac{dy}{dx} \right)$

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